

Understanding the ROC curve in three visual steps



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Published in Towards Data Science

4 min read · Jun 12, 2020



One of the concepts that took me longer to understand in Data Science was the Receiver Operating Characteristic (ROC) curve. This is a visual representation of how well your classification model works.

In this blog, I want to explain how the ROC curve is constructed from scratch in three visual steps.

Step 1: Getting classification model predictions

When we train a classification model, we get the probability of getting a result. In this case, our example will be *the likelihood of repaying a loan*.

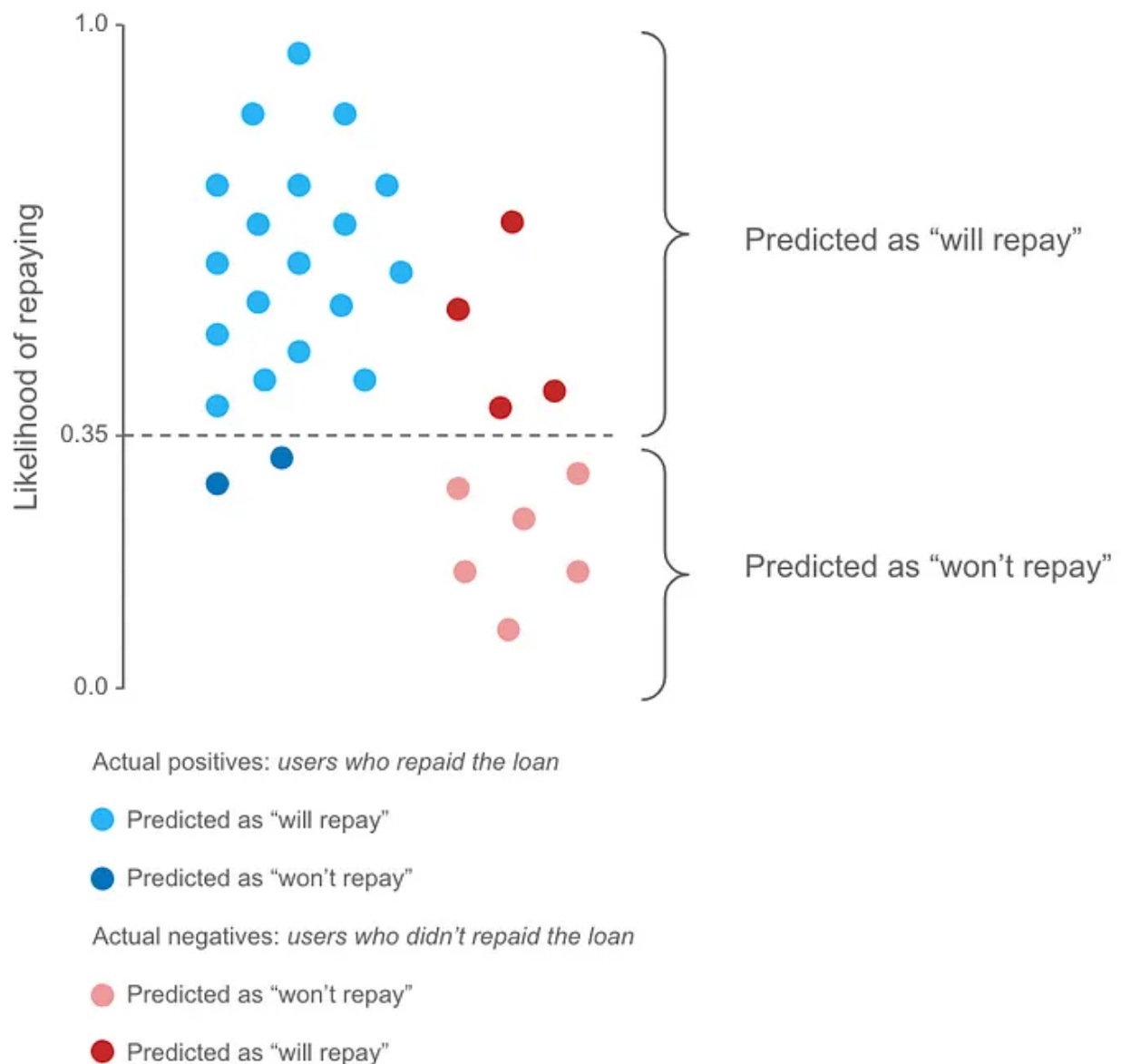


Fig 1: Classification model example

The probabilities range between 0 and 1. The higher the value, the more likely the person is to repay a loan.

Next step is to find a threshold to classify the probabilities as *"will repay"* or *"won't repay"*.

In the example on Fig 1, we've selected a *threshold at 0.35*:

- All predictions at or above this threshold, are classified as *"will repay"*
- All predictions below this threshold, are classified as *"won't repay"*

We then look at which of these predictions were correctly classified or misclassified. With such information, we can build a confusion matrix.

All actual positives, those who *did repay*, are the blue dots.

- If they were classified as “*will repay*”, we have a True Positive (TP)
- If they were classified as “*won’t repay*”, we have a False Negative (FN)

All actual negatives, those who *didn’t repay*, are the red dots.

- If they were classified as “*won’t repay*”, we have a True Negative (TN)
- If they were classified as “*will repay*”, we have a False Positive (FP)

In Fig 2, we can see a summary:

		Actuals	
		Positive	Negative
Predicted	Positive	● True Positive (TP) <i>Predicted as “will repay” and “did repay” in the past</i>	● False Positive (FP) <i>Predicted as “will repay”, but “didn’t repay” in the past</i>
	Negative	● False Negative (FN) <i>Predicted as “won’t repay”, but “did repay” in the past</i>	● True Negative (TN) <i>Predicted as “won’t repay” and “didn’t repay” in the past</i>

Fig 2: Confusion matrix

Step 2: Calculate the True Positive Rate and False Positive Rate

By now, we’ve classified all predictions and we know if the classifications are correct or not. With this, we’ll calculate the next two metrics:

$$\begin{aligned}
 \text{True Positive Rate (TPR)} &= \frac{TP}{P} = \frac{TP}{TP + FN} \\
 \text{also called sensitivity/recall/hit rate} & \\
 \\
 \text{False Positive Rate (FPR)} &= \frac{FP}{N} = \frac{FP}{FP + TN} \\
 \text{also called fall out} &
 \end{aligned}$$

Fig 3: calculations for TPR and FPR

- True positive rate (TPR): *from all people who “did repay” in the past, what percentage did we classify correctly*
- False positive rate (FPR): *from all people who “didn’t repay” in the past, what percentage did we miss-classify*

We can see in Fig 3 the formulas for these calculations.

Below, in the second graph of Fig 4, we can see our original example at the threshold of 0.35. At this point, we

- classified correctly 90% of all positives, those who “paid back” (TPR)
- miss-classified 40% of all negatives, those who “didn’t pay back” (FPR)

We can notice that the results for the TPR and FPR decrease as the threshold gets larger. If we look at the first one, where the threshold is at 0:

- All positives were correctly classified, therefore TPR = 100%
- All negatives were miss-classified, hence FPR = 100%

In the last graph example, where the threshold is at 1:

- All positives were miss-classified, therefore TPR = 0%
- All negatives were correctly classified, hence FPR = 0%

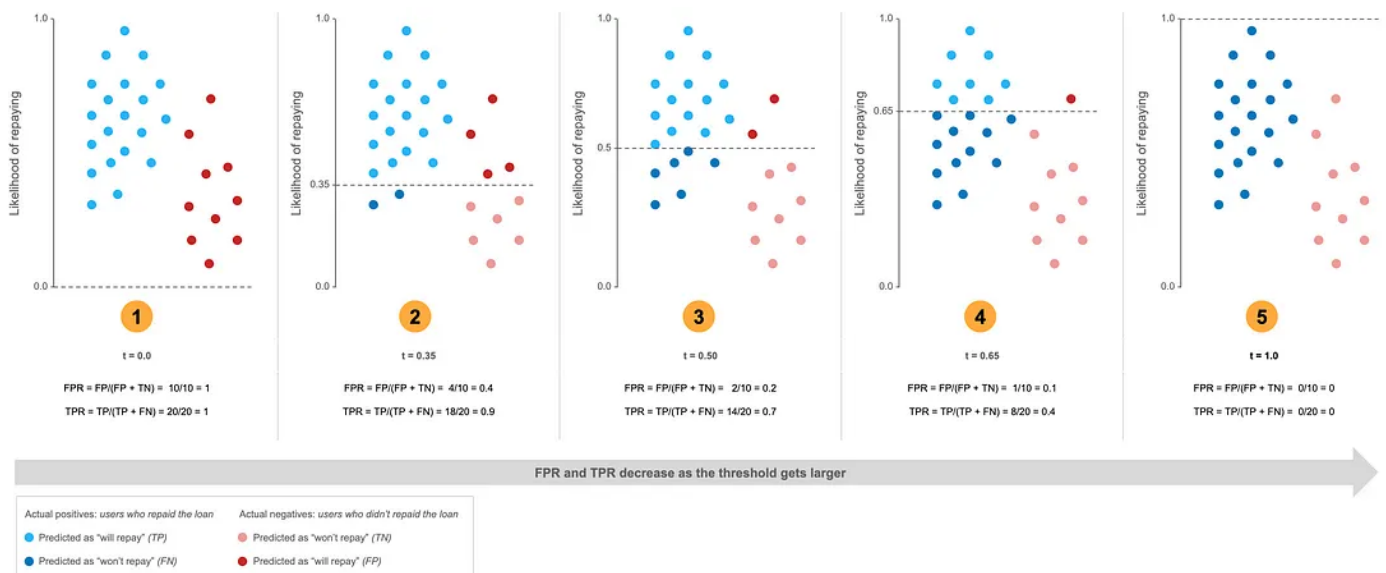


Fig 4: FPR and TPR results for different threshold cut-offs

Overall, we can see this is a trade-off. As we increase our threshold, we'll be better at classifying negatives, but this is at the expense of miss-classifying more positives

Step 3: Plot the the TPR and FPR for every cut-off

To plot the ROC curve, we need to calculate the TPR and FPR for many different thresholds (This step is included in all relevant libraries as `scikit-learn`).

For each threshold, we plot the FPR value in the x-axis and the TPR value in the y-axis. We then join the dots with a line. *That's it!*

Below in Fig 5, we can see how each point of the ROC curve represents the FPR and TRP of a classification at a given cut-off.

Notice how the threshold at 1 leads to the first point at (0, 0) and the threshold at 0 leads to the last point at (1, 1).

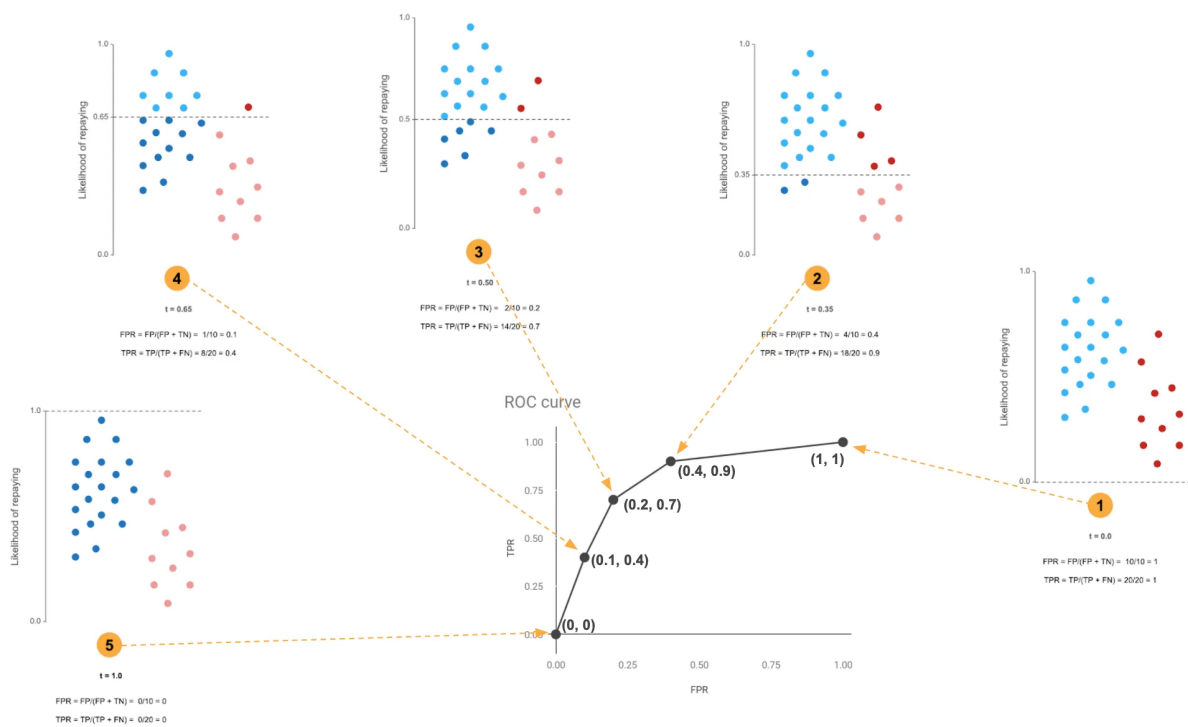


Fig 5: ROC curve example

The area covered below the line is called “*Area Under the Curve (AUC)*”. This is used to evaluate the performance of a classification model. The higher the AUC, the better the model is at distinguishing between classes.

That means that in an ideal world, we’d like to see our line cover most of the upper left of the graph to get a higher AUC.

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